Entity Relationship Diagram & Data Model

Database ER Diagram

```
erDiagram
    companies ||--o{ users : "has many"
    companies ||--o{ employees : "has many"
    companies ||--o{ attendance : "has many"
    companies ||--o{ leave_types : "has many"
    companies ||--o{ leave_requests : "has many"
    companies ||--o{ salary_structures : "has many"
    companies ||--o{ payroll_records : "has many"

    roles ||--o{ users : "assigned to"
    roles ||--o{ role_permissions : "has many"
    permissions ||--o{ role_permissions : "granted through"

    users ||--o| employees : "may have profile"
    users ||--o{ leave_requests : "approves"

    employees ||--o{ attendance : "records"
    employees ||--o{ leave_requests : "submits"
    employees ||--o{ salary_structures : "has"
    employees ||--o{ payroll_records : "receives"

    leave_types ||--o{ leave_requests : "categorizes"
    salary_structures ||--o{ payroll_records : "basis for"

    companies {
        int id PK
        string name
        string registration_number UK
        string email
        string phone
        text address
        string city
        string state
        string country
        string postal_code
        string website
        string industry
        enum company_size
        string logo_path
        string logo_url
        enum status
        enum subscription_plan
        date subscription_expires
        timestamp created_at
        timestamp updated_at
    }

    roles {
        int id PK
        string name UK
        string description
        timestamp created_at
    }

    permissions {
        int id PK
        string name UK
        string module
        string description
        timestamp created_at
    }
```

```
role_permissions {
    int id PK
    int role_id FK
    int permission_id FK
}

users {
    int id PK
    int company_id FK
    int role_id FK
    string email UK
    string password_hash
    enum status
    timestamp last_login
    timestamp created_at
    timestamp updated_at
}

employees {
    int id PK
    int company_id FK
    int user_id FK
    string employee_code
    string first_name
    string last_name
    string email
    string phone
    date date_of_birth
    enum gender
    text address
    date hire_date
    date termination_date
    string department
    string designation
    enum employment_type
    enum status
    timestamp created_at
    timestamp updated_at
}

attendance {
    int id PK
    int company_id FK
    int employee_id FK
    date attendance_date
    time clock_in_time
    time clock_out_time
    decimal total_hours
    enum status
    text notes
    timestamp created_at
    timestamp updated_at
}

leave_types {
    int id PK
    int company_id FK
    string name
    int annual_allocation
    boolean is_paid
    boolean is_active
    timestamp created_at
    timestamp updated_at
}

leave_requests {
    int id PK
    int company_id FK
    int employee_id FK
```

```
        int leave_type_id FK
        date start_date
        date end_date
        int total_days
        text reason
        enum status
        int approver_id FK
        timestamp approval_date
        text rejection_reason
        timestamp created_at
        timestamp updated_at
    }

    salary_structures {
        int id PK
        int company_id FK
        int employee_id FK
        decimal basic_salary
        decimal housing_allowance
        decimal transport_allowance
        decimal other_allowances
        decimal tax_deduction
        decimal insurance_deduction
        decimal other_deductions
        date effective_date
        boolean is_current
        timestamp created_at
        timestamp updated_at
    }

    payroll_records {
        int id PK
        int company_id FK
        int employee_id FK
        int salary_structure_id FK
        int year
        int month
        decimal gross_salary
        decimal total_deductions
        decimal net_salary
        date payment_date
        enum payment_status
        string payment_reference
        timestamp created_at
        timestamp updated_at
    }
```

Data Model Statistics

Table Relationships Summary

Table	Primary Relations	Foreign Keys	Indexes	Constraints
companies	Root entity	0	3	2 unique
roles	Reference table	0	1	1 unique
permissions	Reference table	0	2	1 unique
role_permissions	Junction table	2	1	1 unique composite
users	Tenant entity	2	3	1 unique
employees	Core entity	2	4	2 unique
attendance	Transaction table	2	4	1 unique composite
leave_types	Configuration table	1	2	1 unique composite
leave_requests	Transaction table	4	6	0
salary_structures	Configuration table	2	5	0
payroll_records	Transaction table	3	5	1 unique composite

Data Volume Projections

Table	Small Company (50 employees)	Medium Company (200 employees)	Large Company (1000 employees)
companies	1 record	1 record	1 record
users	50 records	200 records	1,000 records
employees	50 records	200 records	1,000 records
attendance	18,250/year	73,000/year	365,000/year
leave_requests	200/year	800/year	4,000/year
salary_structures	50-100 records	200-400 records	1,000-2,000 records
payroll_records	600/year	2,400/year	12,000/year

Storage Requirements

Table	Record Size	Small Company	Medium Company	Large Company
companies	2KB	2KB	2KB	2KB
users	500 bytes	25KB	100KB	500KB
employees	1.5KB	75KB	300KB	1.5MB
attendance	300 bytes	5.5MB/year	22MB/year	110MB/year
leave_requests	800 bytes	160KB/year	640KB/year	3.2MB/year
salary_structures	400 bytes	20-40KB	80-160KB	400-800KB
payroll_records	600 bytes	360KB/year	1.4MB/year	7.2MB/year
Total Estimated	-	6MB/year	24MB/year	120MB/year

Data Integrity Constraints

Primary Key Constraints

- **11 tables** with auto-incrementing integer primary keys
- **Unique identification** for all entities
- **Referential integrity** maintained across all relationships

Foreign Key Constraints

- **15 foreign key relationships** ensuring data consistency
- **Cascade deletion** for tenant data (company → users → employees)
- **Restrict deletion** for reference data (roles, permissions, leave types)
- **Set null** for optional relationships (user_id in employees)

Unique Constraints

- **8 unique constraints** preventing duplicate data
- **Composite unique keys** for business rules (employee_code per company)
- **Natural unique keys** for business identifiers (email, registration_number)

Check Constraints

- **Month validation** in payroll_records (1-12)
- **ENUM constraints** for controlled vocabularies
- **Data type constraints** ensuring proper data formats

Multi-Tenant Architecture

Tenant Isolation Strategy

- **company_id** in all tenant-specific tables
- **Row-level security** through application middleware
- **Complete data separation** between companies
- **Cascade deletion** for clean tenant removal

Tenant Tables (7 tables)

1. **users** - User accounts per company
2. **employees** - Employee profiles per company
3. **attendance** - Daily attendance records per company
4. **leave_types** - Company-specific leave policies
5. **leave_requests** - Leave applications per company

6. **salary_structures** - Compensation plans per company
7. **payroll_records** - Monthly payroll per company

Shared Reference Tables (4 tables)

1. **companies** - Tenant root entities
2. **roles** - System-wide user roles
3. **permissions** - System-wide permission definitions
4. **role_permissions** - Role-permission mappings

Index Strategy

Performance Indexes

Table	Index Name	Columns	Purpose
companies	idx_companies_status	status	Filter active companies
companies	idx_companies_industry	industry	Group by industry
users	idx_users_company	company_id	Tenant queries
users	idx_users_status	status	Filter active users
employees	idx_employees_company	company_id	Tenant queries
employees	idx_employees_status	status	Filter active employees
employees	idx_employees_department	department	Department reports
attendance	idx_attendance_company	company_id	Tenant queries
attendance	idx_attendance_date	attendance_date	Date range queries
attendance	idx_attendance_status	status	Status filtering
leave_requests	idx_leave_requests_company	company_id	Tenant queries
leave_requests	idx_leave_requests_employee	employee_id	Employee history
leave_requests	idx_leave_requests_status	status	Status filtering
leave_requests	idx_leave_requests_dates	start_date, end_date	Date range queries
payroll_records	idx_payroll_company	company_id	Tenant queries
payroll_records	idx_payroll_period	year, month	Period queries

Composite Indexes (Recommended)

Table	Suggested Index	Columns	Performance Gain
employees	idx_company_status_name	company_id, status, first_name, last_name	50-70%
attendance	idx_company_date_employee	company_id, attendance_date, employee_id	40-60%
leave_requests	idx_company_status_date	company_id, status, start_date	30-50%
payroll_records	idx_company_period_employee	company_id, year, month, employee_id	40-60%

Data Security & Compliance

Sensitive Data Fields

Table	Sensitive Fields	Protection Method
users	password_hash	bcrypt hashing
employees	email, phone, address, date_of_birth	Input validation
salary_structures	All salary fields	Access control
payroll_records	All financial fields	Access control

Audit Trail

- **created_at** timestamp on all tables
- **updated_at** timestamp with auto-update
- **Immutable records** for payroll and attendance
- **Soft delete capability** for critical data

GDPR Compliance Features

- **Data minimization** - only necessary fields collected
- **Purpose limitation** - clear data usage definitions
- **Storage limitation** - automatic retention policies possible

- **Data portability** - JSON export capability
- **Right to erasure** - cascade deletion support

This ER diagram and data model documentation provides a comprehensive view of the database structure, relationships, and capacity planning for the multi-tenant HRMS system.